Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a hybrid AI architecture that combines the parametric knowledge of large language models with non-parametric, external knowledge sources. Rather than relying solely on information encoded in model weights during training, RAG systems dynamically retrieve relevant documents at inference time and condition the generation on that retrieved context. This approach addresses key limitations of standalone LLMs: knowledge staleness, hallucination, and the inability to cite sources.

Vector Databases and Semantic Search

At the core of any RAG pipeline lies a vector database. Documents are split into chunks, each chunk is encoded into a dense embedding vector using a transformer encoder model such as sentence-transformers. These vectors capture semantic meaning: two chunks about similar topics will have embeddings close together in high-dimensional space. FAISS (Facebook AI Similarity Search) is a widely used open-source library for efficient approximate nearest-neighbour search over millions of vectors. It supports several index types: Flat (exact, slow), IVF (inverted file, faster), and HNSW (graph-based, fastest for large corpora).

Chunking Strategies

How a document is split into chunks significantly impacts retrieval quality. Fixed-size character splitting is simple but cuts across sentence and paragraph boundaries. Recursive character splitting uses a priority list of separators to find natural split points. Semantic chunking groups sentences by embedding similarity rather than character count, producing topically coherent chunks at the cost of higher preprocessing time. Chunk size is a key hyperparameter: larger chunks give more context per retrieval but reduce precision; smaller chunks are more precise but may lack context for generation.

Re-ranking and Maximal Marginal Relevance

First-stage retrieval with dense embeddings optimises for relevance but can return redundant chunks that all say the same thing. Maximal Marginal Relevance (MMR) selects chunks iteratively: each selected chunk must be relevant to the query but dissimilar to already-selected chunks. Cross-encoder re-ranking uses a separate model that takes a query-chunk pair as input and outputs a fine-grained relevance score. Cross-encoders are more accurate than bi-encoders for ranking but cannot be used for first-stage retrieval.

Evaluation Metrics for RAG Systems

Evaluating RAG systems requires measuring both retrieval quality and generation quality. For retrieval: Recall@K measures what fraction of ground-truth relevant documents appear in the top K results; MRR rewards systems that rank the correct document higher. For generation: RAGAS provides reference-free metrics including faithfulness, answer relevancy, and context precision. Latency is also critical: end-to-end query latency should stay under 3 seconds for interactive applications.